

PRACTICE SHEET-1

1. What type of classification is needed to enumerate the female population of India?
(1) Geographical (2) Chronological
(3) Qualitative (4) Quantitative
2. For less than ogive, the cumulative frequencies are plotted against which of the following?
(a) Upper limits of class intervals
(b) Lower limits of the class intervals
(c) Mid-points of the class intervals
(d) Either (b) or (c)
3. The cumulative frequency curve of a frequency distribution with 6 classes and total frequency 60 is a straight line. Consider the following statements:
I. The first and the last classes have a frequency of 10 each.
II. Both the middle classes have a total frequency of 30.
III. The frequency distribution does not have a mode.
Which of the above statements are incorrect?
(a) I and II (b) I and III
(c) II and III (d) I, II and III
4. In Statistics, a suitable graph for representing the partitioning of total into subparts is
(a) An ogive (b) A pictograph
(c) A histogram (d) A pie chart
5. Which one of the following statements is not correct with reference to a histogram?
(a) Frequency curve is obtained by joining the mid-points of the top of the adjacent rectangles with smooth curves
(b) Histogram is drawn for continuous data
(c) The height of the bar is proportional to the frequency of that class
(d) Mode of the distribution can be obtained from the histogram
6. Assume that population densities of 5 major states of India are given. Which one of the following diagrams is suitable to represent the data?
(a) Single bar diagram
(b) Percentage bar diagram
(c) Pie diagram
(d) Since population density is a ratio, it cannot be represented by any diagram
7. Which one of the following represents statistical data?
(a) The names of all owners of shops located in a shopping complex
(b) A list giving the names of all states of India
(c) A list of all European countries and their respective capital cities
(d) The volume of a rainfall in certain geographical area, recorded every month for 24 consecutive months
8. Data on percentage distribution of area of land in acres owned by households in two districts of a particular state are as follows:

Land Holding	District A	District B
0.01 - 0.99	5.62	13.53
1.0 - 2.49	18.35	21.84
2.5 - 7.49	47.12	39.32
7.5 - 12.49	19.34	12.15
12.5 - 19.99	7.21	7.43
20.0 - 29.99	2.36	5.73

What is the appropriate diagram to represent the above data?
(a) Pie diagram
(b) Histogram
(c) Bar chart
(d) None of the above
9. Examples of data are given below:
I. Information on households collected by an investigator by door to door visits.
II. Data on the percentage of literates, sexwise, for the different districts of a state collected from records of the census of India.
III. General information about families, collected by telephonic interviews.
Which one of the following in respect of the above is correct?
(a) I and II are primary data
(b) I and III are primary data
(c) II and III are primary data
(d) I, II and III are primary data
10. Consider the following statements in respect of histogram.
I. Histogram is an equivalent graphical representation of the frequency distribution.

II. Histogram is suitable for continuous random variables, where the total frequency of an interval is evenly distributed over the interval.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

11. Consider the following statements in respect of a histogram:

I. The histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars.

II. The height of the rectangle is determined by the frequency of the class it represents.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

12. Consider the following statements in respect of class intervals of grouped frequency distribution.

1. Class intervals need not be mutually exclusive.

2. Class intervals should be exhaustive

3. Class intervals need not be of equal width

Which of the above statements are correct?

- (a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3

13. Consider the following statements in respect of a histogram.

1. The total area of the rectangles in a histogram is equal to the total area bounded by the corresponding frequency polygon and the axis.

2. When class intervals are unequal in a frequency distribution, the area of the rectangle is proportional to the frequency.

Which of the above statements is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor

14. Consider the following statements in respect of histogram.

I. The histogram is a suitable representation of a frequency distribution of a continuous variable.

II. The area included under the whole histogram is the total frequency.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

15. The cumulative frequency of the largest observed value must always be:

- (a) Less than the total number of observations
(b) Greater than the total number of observations
(c) Equal to total number of observations
(d) Equal to mid-point of the last class interval

16. Consider the following statements:

I. A continuous random variable can take all values in an interval.

II. A random variable which takes a finite number of values is necessary discrete.

III. Construction of a frequency distribution is based on data which are discrete.

Which of the above statements are correct?

- (a) Both I and II (b) Both II and III
(c) Both I and III (d) All of these

17. Consider the following statements in respect of histogram.

I. Pie diagram is suitable for categorical data.

II. The arc length of a sector of a pie diagram is proportional to the value of the component represented by the sector.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

18. A Histogram is a graphical presentation of a frequency distribution of a

- (1) individual series (2) discrete series
(3) continuous series (4) none of these

19. Which of the following diagram is drawn by joining mid-points of the tops of all rectangles in a histogram:

- (1) frequency distribution
(2) frequency polygon
(3) frequency curve
(4) none of these

20. What is the shape of 'less than ogive'

- (1) Rising upward (2) falling downward
(3) Parallel to X-axis (4) Parallel to Y axis

21. Which of the following is a shape of frequency distribution curve?

- (1) A-shaped (2) B-shaped
(3) U shaped (4) All the above

22. Normal curves are also known as:

- (1) J - shaped

(2) L - shaped

(3) U - shaped

(4) bell-shaped curves

23. In a pie diagram, there are four slices with angles 150° , 90° , 60° and 60° . A new pie diagram is formed by deleting one of the slices having angle 60° in the given pie diagram. In the new pie diagram

(a) The largest slice has angle 150°

(b) The smallest slice has angle 70°

(c) The largest slice has angle 180°

(d) The smallest slice has angle 90°

24. Which one of the following is not correct?

The proportion of various items in a pie diagram is proportional to the

(a) Areas of slices

(b) Angles of slices

(c) Lengths of the curved arcs of the slices

(d) Perimeters of the slices

25. Consider the following statements in respect of histogram.

I. Primary data involve more time and more expense.

II. Investigator is a person who plans and conducts an empirical investigation independently.

Which of the above statement (s) is/are correct?

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither 1 nor 2

26. Consider the following statements in respect of histogram.

I. Positively skewed curves have their tail more spread towards right.

II. Histogram is drawn only for equal class intervals.

Which of the above statement (s) is/are correct?

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither 1 nor 2

27. Data can be represented in which of the following forms?

1. Textual form

2. Tabular form

3. Graphical form

Select the correct answer using the codes given below:

(a) 1 and 2

(b) 2 and 3

(c) 1 and 3

(d) 1, 2 and 3

28. Which one of the following can be obtained from an ogive?

(a) Mean

(b) Median

(c) Geometric

(d) Mode

29. A set of annual numerical data, comparable over the years, is given for the last 12yr.

Consider the following statements.

I. The data is best represented by a broken line graph, each corner (turning point) representing the data of one year.

II. Such a graph depicts the chronological change and also enables one to make a short-term forecast.

Select the correct answer using the codes given below:

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither I nor II

30. Which term is not related with statistics?

(a) Data

(b) Histogram

(c) Domain

(d) Range

ANSWER KEY

1.	(4)	2.	(1)	3.	(2)	4.	(4)	5.	(4)	6.	(3)	7.	(4)	8.	(3)	9.	(2)	10.	(3)
11.	(3)	12.	(1)	13.	(3)	14.	(1)	15.	(3)	16.	(1)	17.	(3)	18.	(3)	19.	(2)	20.	(1)
21.	(3)	22.	(4)	23.	(3)	24.	(4)	25.	(3)	26.	(1)	27.	(4)	28.	(2)	29.	(3)	30.	(3)

PRACTICE SHEET-2

Direction for next three:

Study the following table and answer the questions.

School	No. of students scoring marks less than 50%	Percentage of students scoring marks more than 50%	No. of students appeared
A	240	55	600
B	220	40	400
C	300	20	375
D	280	10	350
E	210	25	300

- The ratio of the total number of students scoring marks less than 50% to that of scoring marks exactly 50% is
(1) 50 : 3 (2) 25 : 2
(3) 25 : 4 (4) 35 : 2
- Which school has the highest number of students scoring exactly 50% marks?
(1) D (2) E
(3) B (4) A
- The total number of students scoring 50% or more marks is
(1) 1250 (2) 875
(3) 775 (4) 675

Direction for next three:

Study the following table and answer the questions.

Source of Income	Schools				
	A	B	C	D	E
Tuition Fee.	120	60	210	90	120
Term Fee	24	12	45	24	30
Donation	54	21	60	51	60
Grants	60	54	120	42	55
Miscellaneous	12	3	15	3	15
Total	270	150	450	210	280

- For school E, what per cent of the income from miscellaneous is the income from donation?
(1) 25% (2) 40%
(3) $\frac{1}{4}$ % (4) 400%
- Which school has the highest percentage of income from tuition fee out of its total income?

- (1) A (2) B
(3) C (4) D

6. In case of how many schools, is the income by way of tuition fee, is less than four times of term fee

- (1) 0 (2) 1
(3) 2 (4) 3

Direction for next three:

Study the following table and answer the questions.

Food Grains Production in a country in 1999 (in lakh tons)					
State	Rice	Wheat	Jowar	Pulses	Others
P	45	103	—	27	29
Q	48	86	73	19	15
R	59	32	67	14	31
S	41	37	59	21	15
T	37	22	41	13	11
U	68	15	12	—	18
V	57	8	7	12	10
W	38	28	31	22	45

- Which State had the highest grain production
(1) P (2) Q
(3) R (4) S
- What was the proportion of rice production to wheat production in the country ?
(1) 1 : 1 (2) 0.8 : 1
(3) 1.2 : 1 (4) 2 : 1
- Jowar was the most important food grain in the State/States :
(1) Q, R, S (2) Q
(3) R, S (4) R, S, T

Direction for next three:

The following table shows the production of food grains (in million tonnes) in a State for the period 1988 to 1992. Read the table and answer the following questions.

Year	Production in million tonnes				Total
	Wheat	Rice	Maize	Other Cereals	
1988	580	170	150	350	1250
1989	600	220	234	400	1454
1990	560	240	228	420	1448
1991	680	300	380	460	1820
1992	860	260	340	500	1960
Total	3280	1190	1332	2130	7932

10. During 1990, the percentage of decrease in production of maize as against the previous year was:

- (1) 2.63% (2) 2.56%
(3) 2.71% (4) 2.47%

11. In 1991, the increase in production over the previous year was maximum for :

- (1) Wheat (2) Rice
(3) Maize (4) Other Cereals

12. The increase in the production of other cereals (over the previous year) was minimum during the year :

- (1) 1989 (2) 1990
(3) 1991 (4) 1992

Direction for next three:

Study the following table and answer the questions.

Years	Men	or Women	Children	Total	Increase (+) decrease (-) over preceding year
1988	65104	60387	—	146947	—
1989	70391	62516	—	—	+ (11630)
1990	—	63143	20314	153922	—
1991	68395	—	21560	—	— (5337)
1992	71274	659935	23789	16098	—

13. The number of children in 1988 is :

- (1) 31236 (2) 125491
(3) 14546 (4) 21456

14. The total population in 1989 is :

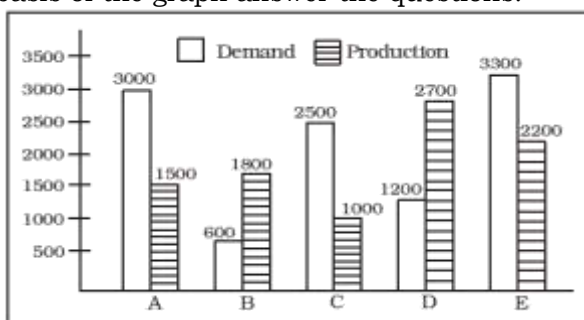
- (1) 144537 (2) 158577
(3) 146947 (4) 149637

15. Number of children in 1989 is :

- (1) 25670 (2) 14040
(3) 13970 (4) 15702

Direction for next three:

The following chart represents Demand and Production for 5 companies ABCDE. On the basis of the graph answer the questions.



16. If company A desires to meet the demand by purchasing surplus production of company, then the most suitable company is :

- (1) C (2) D
(3) E (4) B

17. If $x\%$ of demand for company C equals demand for company B, then x equals

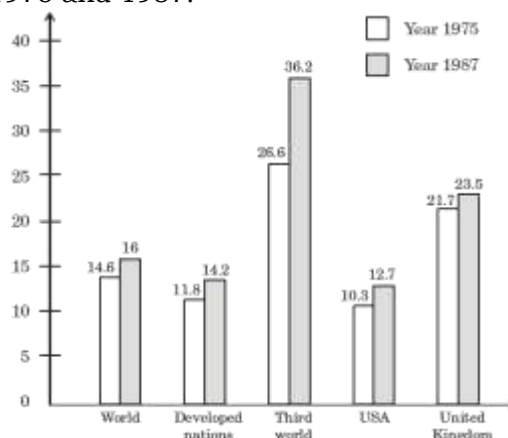
- (1) 24 (2) 20
(3) 60 (4) 4

18. If the production of company D is h times of the production of company A. Then h equals :

- (1) 1.5 (2) 2.5
(3) 1.2 (4) 1.8

Direction for next three:

Study the following Bar graph and answer the questions. The Bar Graph gives the annual rates of inflation in percentages for 1975 and 1987.



19. From 1975 to 1987, inflation rate increased in the third world countries approximately by

- (1) 10% (2) 20%
(3) 30% (4) 36%

20. The change in rate of inflation was least in which of the following?

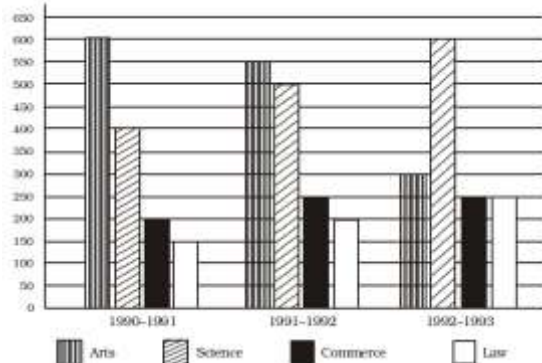
- (1) Developed Nations (2) United Kingdom
(3) World (4) Third world

21. Comparing the figures for USA vis-a-vis the developed nations, it can be concluded that.

- (1) USA had better control on inflation
(2) Developed nations had better control on inflation
(3) The inflation rate continues to be the same for USA and developed nations
(4) No conclusions can be drawn

Direction for next three:

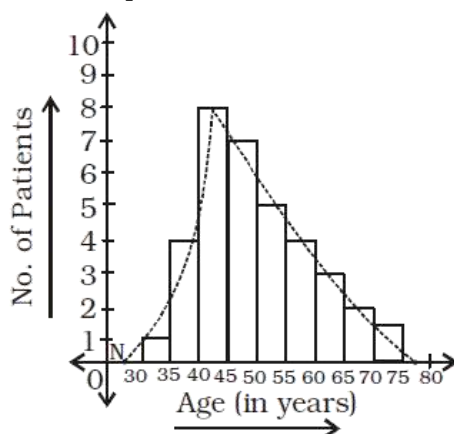
Shown below is the multiple bar diagram depicting the changes in the student's strength of a college in four faculties from 1990-91 to 1992-93. (Scale 1 cm = 100)



- 22.** How much per cent was the increase in science students in 1992-93 over 1990-91?
 (1) 50% (2) 150%
 (3) 66.66% (4) 75%
- 23.** The respective ratio of the number of commerce students to the number of law students from 1990-91 to 1992-93 is :
 (1) 6 : 7 (2) 7 : 6
 (3) 14 : 1 (4) 2 : 8
- 24.** During which year the strength of arts faculty was minimum?
 (1) 1990-91 (2) 1991-92
 (3) 1992-93 (4) None of these

Direction for next Four:

The diagram shows the age-distribution of the patients admitted to a hospital on a particular day. Study the diagram and answer the questions.

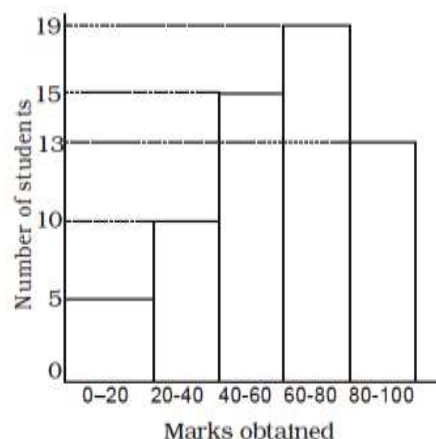


- 25.** Total number of patients of age more than 55 years, who got admitted to the hospital is
 (1) 4 (2) 7
 (3) 9 (4) 10

- 26.** Number of patients of age more than 40 years and less than 55 years, who got admitted to the hospital on that day is
 (1) 20 (2) 30
 (3) 15 (4) 12
- 27.** Percentage of patients of age less than 45 years, who got admitted to the hospital on that day is approximately equal to
 (1) 14% (2) 20%
 (3) 37% (4) 62%
- 28.** About 11% of the patients who got admitted to the hospital on that particular day were of age
 (1) either between 35 years and 40 years or between 55 years and 60 years
 (2) between 60 years and 65 years
 (3) between 35 years and 40 years
 (4) between 35 years and 40 years and between 55 years and 60 years.

Direction for next two:

In each of the following questions, the following histogram shows the relationship between the marks obtained by the students and the number of students in an examination. Study the histogram and answer the questions.



- 29.** The percentage of students who obtained 40 marks or less is
 (1) 25
 (2) less than 25, but not 15
 (3) more than 24
 (4) 15
- 30.** The ratio of the number of students who obtained 60 or more marks to that of students who obtained 60 or less marks is

- (1) 15 : 16 (2) 15 : 19
(3) 14 : 17 (4) 16 : 15

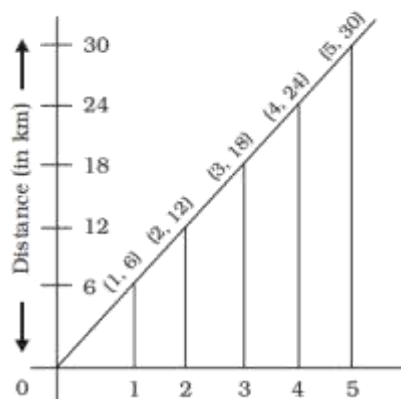
ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(1)	5.	(3)	6.	(2)	7.	(2)	8.	(3)	9.	(4)	10.	(2)
11.	(3)	12.	(1)	13.	(4)	14.	(2)	15.	(1)	16.	(2)	17.	(1)	18.	(4)	19.	(4)	20.	(2)
21.	(1)	22.	(1)	23.	(2)	24.	(3)	25.	(4)	26.	(1)	27.	(3)	28.	(4)	29.	(2)	30.	(1)

PRACTICE SHEET-3

Direction for next three:

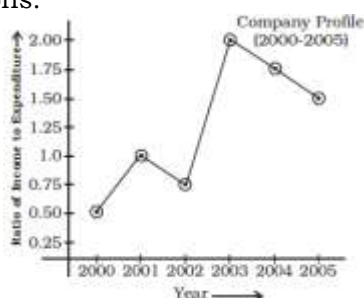
The graph given here shows a car following the linear path with uniform speed. Study the graph and answer the questions.



- The speed of the car is
(1) 12 km/hr (2) 6 km/hr
(3) 18 km/hr (4) 24 km/hr
- The speed of the car (in metres per minute) is
(1) 60 (2) 100
(3) 600 (4) 1000
- The distance travelled by the car in 4.5 hours is
(1) 27 km (2) 30 km
(3) 36 km (4) 40 km

Direction for next two:

Study the following graph and answer the questions.



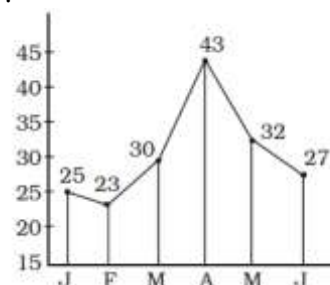
- Find the percentage decrease in income from 2001 to 2002.

- (1) 50% (2) 33%
(3) 37.5% (4) Data inadequate

- If the income shows positive growth every year throughout the period (2000 - 2005), then in how many years the expenditure shows a positive growth?

- (1) 5 (2) 3
(3) 4 (4) 2

- Given is a line graph showing the number of accidents in a city during the first 6 months of 1999.

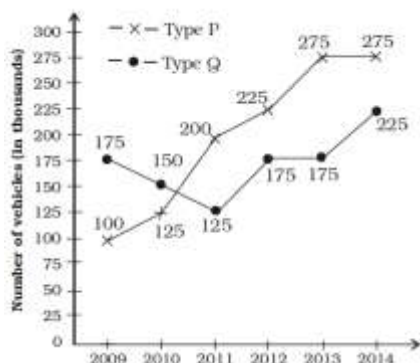


The decrease % of accidents from May to June is

- (1) 15.375% (2) 15.125%
(3) 15.625% (4) 15.875%

Direction for next three:

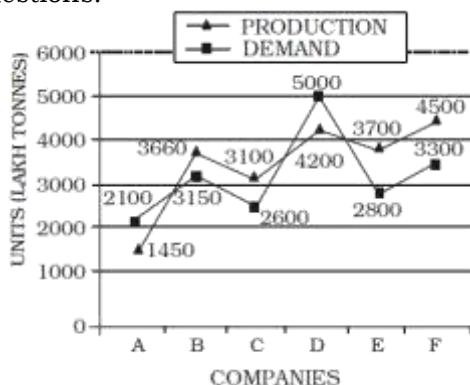
The following graph shows production (in thousands) of two types (P and Q) of vehicles by a factory over the years 2009 to 2014. Study the graph and answer the given questions.



7. In how many of the given years, was the production of Type P vehicles of the company more than the average production of this type vehicles in the given years?
(1) 3 (2) 4
(3) 2 (4) 5
8. Approximate percentage decrease in production of Type Q vehicles from 2010 to 2011 is
(1) 10.1 (2) 16.7
(3) 14.3 (4) 12.5
9. The total production of Type P vehicles in the years 2009 and 2011 is what percent of total production of Type Q vehicles in 2010 and 2014?
(1) 75 (2) 69.25
(3) 80 (4) 81.25

Direction for next three:

In the following questions, the Graph shows the demand and production of different companies. Study the graph and answer the questions.



10. The demand of company B is what percentage of the production of company F?
(1) 60% (2) 70%
(3) 80% (4) 50%

11. What is the difference between the average demand and the average production of the companies (in lakh tonnes)? [Approximately]

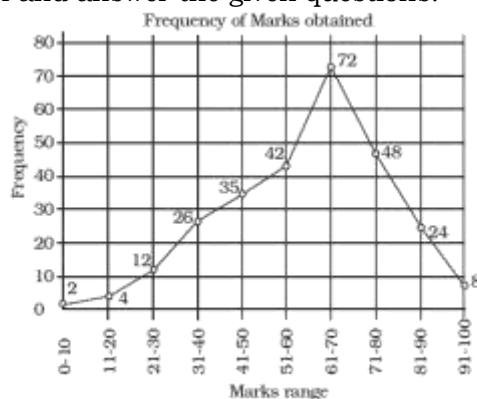
(1) 200 (2) 325
(3) 275 (4) 250

12. The production of company A is approximately what percent of the demand of company C?

(1) 50% (2) 65%
(3) 60% (4) 55%

Direction for next three:

The marks obtained by 273 examinees are shown by the frequency polygon. Given that mean marks is 59.5. Study the frequency polygon and answer the given questions.



13. The number of examinees getting more than average marks is

(1) 72 (2) 105
(3) 152 (4) 164

14. Percentage of the students who get above 80% marks is

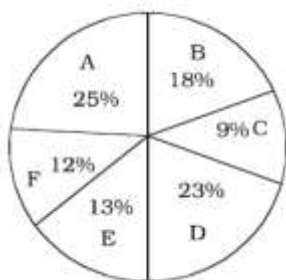
(1) 9.81 (2) 10.53
(3) 11.28 (4) 11.72

15. Percentage of the students who got marks above 60% and below 80% is

(1) 43.95 (2) 48.39
(3) 51.06 (4) 56.84

Direction for next three:

The Pie chart given here represents the domestic expenditure of a family in per cent. Study the chart and answer the following questions if the total monthly income of the family is 33,650.

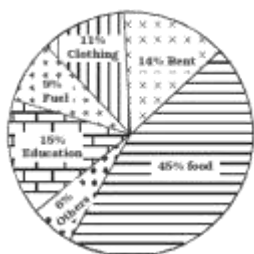


A:Expenditure on food
B:Expenditure on house-rent
C:Expenditure on entertainment
D:Expenditure on education and maintenance of children
E:Medical and miscellaneous expenditure
F:Deductions towards provident fund

16. The house rent per month is :
(1) 6000 (2) 6152
(3) 6057 (4) 6048
17. The annual savings in the form of provident fund would be
(1) 48,456 (2) 48,540
(3) 44,856 (4) 45,480
18. After provident fund deductions and payment of house rent, the total monthly income of the family remains
(1) 23,545 (2) 24,435
(3) 23,555 (4) 25, 355

Direction for next three:

The pie chart given below shows the spending of a family on various heads during a month. Study the graph and answer the following questions.



19. If the total income of the family is 25,000, then the amount spent on Rent and Food together is
(1) 17,250 (2) 14, 750
(3) 11, 250 (4) 8, 500
20. What is the ratio of the expenses on Education to the expenses on Food?

- (1) 1 : 3 (2) 3 : 1
(3) 3 : 5 (4) 5 : 3

21. Expenditure on Rent is what percent of expenditure on Fuel?

- (1) 135% (2) 156%
(3) 167% (4) 172%

Direction for next three:

Study the pie chart and answer questions.



22. The ratio of the total amount spent for food and medicine is

- (1) 1 : 2 (2) 3 : 1
(3) 12 : 5 (4) 11 : 5

23. If the total amount spent on the family during the year was 50,000, how much they spent for buying clothes?

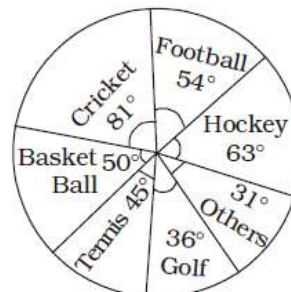
- (1) 6,250 (2) 6,500
(3) 7,250 (4) 7,500

24. If the total amount spent on the family during the year was 35,000, the amount spent for study of children and food together was

- (1) 19,250 (2) 19,500
(3) 19,750 (4) 19,850

Direction for next three:

The Pie Chart shows the expenditure of a country on various sports during a particular year. Study the graph and answer the questions



25. If the total amount spent on cricket and hockey together is 80,000, the total amount spent on sports is

- (1) 1,00,000 (2) 2,00,000
(3) 2,50,000 (4) 3,00,000

26. How much per cent more is spent on Hockey than that on Golf?

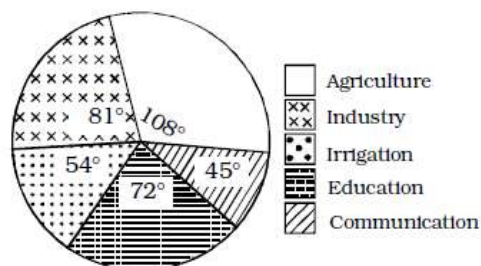
- (1) 27% (2) 35%
(3) 37.5% (4) 75%

27. How much per cent less is spent on football than that on cricket?

- (1) 22.22% (2) 27%
(3) 33.33% (4) 37.5%

Direction for next three:

The pie-chart shows the proposed outlay for different sectors during a Five-Year plan of Government of India. Total outlay is Rs. 40,000 crores. By reading the pie-chart answer the following three questions.



28. What is the proposed outlay for Education?

- (1) Rs. 6000 crores (2) Rs. 8000 crores
(3) Rs. 9000 crores (4) Rs. 7000 crores

29. If the proposed outlay of Irrigation is $x\%$ of the proposed outlay of Agriculture, then x is equal to

- (1) 50% (2) 15%
(3) 25% (4) 75%

30. What is the ratio between the proposed outlay of Irrigation and Communication?

- (1) 9 : 8 (2) 3 : 2
(3) 9 : 5 (4) 6 : 5

ANSWER KEY

1.	(2)	2.	(2)	3.	(1)	4.	(4)	5.	(4)	6.	(3)	7.	(1)	8.	(2)	9.	(4)	10.	(2)
11.	(3)	12.	(4)	13.	(3)	14.	(4)	15.	(1)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(1)
21.	(2)	22.	(3)	23.	(1)	24.	(1)	25.	(2)	26.	(4)	27.	(3)	28.	(2)	29.	(1)	30.	(4)

PRACTICE SHEET-4

1. The median of the observations 22, 24, 33, 37, $x + 1$, $x + 3$, 46, 47, 57, 58 in ascending order is 42. What are the values of 5th and 6th observations respectively?

- (a) 42, 45 (b) 41, 43
(c) 43, 46 (d) 40, 40

2. The class marks in a frequency table are given to be 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. The class limits of the first five classes are

- (a) 3-7, 7-13, 13-17, 17-23, 23-27
(b) 2.5-7.5, 7.5-12.5, 12.5-17.5, 17.5-22.5, 22.5-27.5
(c) 1.5-8.5, 8.5-11.5, 11.5-18.5, 18.5-21.5, 21.5-28.5
(d) 2-8, 8-12, 12-18, 18-22, 22-28

3. Consider the following discrete frequency distribution:

x	1	2	3	4	5	6	7	8
f	3	15	45	57	50	36	25	9

What is the value of median of the distribution?

- (a) 4 (b) 5
(c) 6 (d) 7

4. The following observation has been arranged in ascending order:

29, 32, 48, 52, x , $x + 3$, 71, 75, 80, 92

If the median of data is 61.5, then what is the value of x ?

- (a) 54 (b) 60
(c) 62 (d) 56

5. What is the geometric means of the observations?

125, 729, 1331?

- (a) 495 (b) 1485
(c) 2221 (d) None of these
6. What is the median of a numbers 4.6, 0, 9.3, -4.8, 7.6, 2.3, 12.7, 3.5, 8.2, 6.1, 3.9, 5.2?
(a) 3.8 (b) 4.9
(c) 5.7 (d) 6.0
7. Consider the following statements:
I.The sum of deviations from mean is always zero.
II.The sum of absolute deviations is minimum when taken around median.
Which of the above statement is/are correct?
(a) I only (b) 2 only
(c) Both I and II (d) Neither I nor II
8. Consider the following statements:
I.Mean is independent of change in scale and changes in origin
II.Variance is independent of change in scale but not in origin.
Which of the above statements is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
9. Let $\bar{x}$ be the mean of $x_1, x_2, x_3, \dots, x_n$. If $x_i = a + cy_i$ for some constants a and c , then what will be the mean of $y_1, y_2, y_3, \dots$,
(a) $a + c\bar{x}$ (b) $a - \frac{1}{c}\bar{x}$
(c) $\frac{1}{c}\bar{x} - a$ (d) $\frac{\bar{x} - a}{c}$
10. If x_1 and x_2 are positive quantities, then the condition for the difference between the arithmetic mean and the geometric mean to be greater than 1 is:
(a) $x_1 + x_2 > 2\sqrt{x_1x_2}$
(b) $\sqrt{x_1} + \sqrt{x_2} > \sqrt{2}$
(c) $|\sqrt{x_1} - \sqrt{x_2}| > \sqrt{2}$
(d) $x_1 + x_2 < 2(\sqrt{x_1x_2} + 1)$

DIRECTIONS (for next two): Read the following information carefully to answer the questions that follow.

The average age of 6 persons living in a house is 23.5 years. Three of them are majors and their average age is 42 years. The difference in ages of the three minor children is same.

11. What is the mean of the ages of minor children?
(a) 3 years (b) 4 years
(c) 5 years (d) 6 years
12. What is the median of the ages of minor children?
(a) 3 years
(b) 5 years
(c) 7 years
(d) Cannot be determined
13. Consider the following statements:
1.The mean and median are equal in symmetric distribution.
2.The range is the difference between the maximum value and the minimum value in the data.
3.The sum of the area of the rectangles in the histogram is equal to the total area bounded by the frequency polygon and that horizontal axis.
Which of the above statements are correct?
(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) All 1, 2 and 3

Direction (for next two): The following table gives the frequency distribution of life length in hours of 100 electric bulbs having median life 20 h.

Life of bulbs (in hours)	Number of bulbs
8-13	7
13-18	x
18-23	40
23-28	y
28-33	10
33-38	2

14. What is the missing frequency 'x'?
(a) 31 (b) 27
(c) 24 (d) 14
15. What is the missing frequency 'y'?
(a) 27 (b) 24
(c) 14 (d) 11
16. A random sample of 20 people is classified in the following table according to their ages:

Age	Frequency
15-25	2
25-35	4
35-45	6
45-50	5
55-65	3

What is the mean age of this group of people?

- (a) 41.0 (b) 41.5
(c) 42.0 (d) 42.5

17. The scores of 15 students in an examination were recorded as 10, 5, 8, 16, 18, 20, 8, 10, 16, 20, 18, 11, 16, 14 and 12. After calculating the mean, median and mode, an error is found. One of the values is wrongly written as 16 instead of 18. Which of the following measures of central tendency will change?

- (a) Mean and median
(b) Median and mode
(c) Only mode
(d) Mean and mode

18. What is the mean deviation from the mean of the numbers 10, 9, 21, 16, 24?

- (a) 5.2 (b) 0.5
(c) 0.6 (d) 0.7

19. For the data 3, 5, 1, 6, 5, 9, 5, 2, 8, 6 the mean, median and mode are x, y and z respectively.

Which one of the following is correct?

- (a) $x = y \neq z$ (b) $x \neq y = z$
(c) $x \neq y \neq z$ (d) $x = y = z$

20. Sets A, B and C contain 5 numbers each. The medians of the numbers in these sets are 3, 8 and 11, respectively. What is the median of the combined 15 numbers of the three sets?

- (a) 8
(b) 7
(c) $22/3$
(d) Cannot be determined

21. The mean of five numbers is 30. If one number is excluded, their mean becomes 28. The excluded number is:

- (a) 28 (b) 30
(c) 35 (d) 38

22. Consider the following types of data:

- I. Marks of students who appeared for a test of 100 marks.
II. Collar sizes of 200 shirts sold in a week.
III. Monthly incomes of 250 employees of a factory. For which of the above data, mode is a suitable measure of central tendency?

- (a) Both I and II (b) Only II
(c) Both I and III (d) I, II and III

Directions (for next three): The number of telephone calls received in 245 successive

one minute intervals at an exchange is given below in the following frequency distribution.

Number	0	1	2	3	4	5	6	7
Frequency	14	21	25	43	51	40	39	12

23. What is the mean of the distribution?

- (a) 3.76 (b) 3.84
(c) 3.96 (d) 4.05

24. What is the median of the distribution?

- (a) 3.5 (b) 4
(c) 4.5 (d) 5

25. What is the mode of the distribution?

- (a) 3 (b) 4
(c) 5 (d) 6

26. The cumulative frequency of the largest observed value must always be:

- (a) Less than the total number of observations
(b) Greater than the total number of observations
(c) Equal to total number of observations
(d) Equal to mid-point of the last class interval

27. Some measure central tendency for n discrete observations are given below:

- I. Arithmetic mean
II. Geometric mean
III. Harmonic mean
IV. Median

A desirable property of a measure of central tendency is, if every observation is multiplied by c, then the measure of central tendency is also multiplied by c, where $c < 0$.

Which of the above measures satisfy the property.

- (a) I, II and III (b) I, II and IV
(c) III and IV (d) I, II, III and IV

Directions (for next two): The following table gives the continuous frequency distribution of a continuous variable X.

Class Interval	Frequency
0-10	5
10-20	10
20-30	20
30-40	5
40-50	10

28. What is the median of the above frequency distributions?

- (a) 23 (b) 24
(c) 25 (d) 26

29. What is the mean of the above frequency distribution?

- (a) 25 (b) 26
(c) 27 (d) 28

30.

Class Interval	1-5	6-10	11-15	16-20
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Frequency	3	7	6	5
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Consider the following statements in respect of the above frequency distribution.

- I. The median is contained in the modal class.
II. The distribution is bell-shaped

Which of the above

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWER KEY

1.	(b)	2.	(c)	3.	(b)	4.	(b)	5.	(a)	6.	(b)	7.	(c)	8.	(d)	9.	(d)	10.	(c)
11.	(c)	12.	(b)	13.	(d)	14.	(b)	15.	(c)	16.	(b)	17.	(d)	18.	(a)	19.	(d)	20.	(d)
21.	(d)	22.	(b)	23.	(a)	24.	(b)	25.	(b)	26.	(c)	27.	(d)	28.	(c)	29.	(b)	30.	(b)

CDS PYQ

1. Which of the following are the examples of discrete variables?

- I. Number of errors per page in a book
II. Height of individuals measured in centimeter
III. Waiting time to failure of electric bulbs
IV. Number of levels on branches of a tree

Select the correct answer using the codes given below

[CDS 2013 (1)]

- (a) Only I (b) I and IV
(c) III and IV (d) II and IV

Directions (for next three): Read the following information carefully to answer the questions that follow:

In a frequency distribution having class intervals 0-10, 10-20, 20-30 and 30-40 the respective frequency are x , $x+8$, $x-2$, and $x-4$ and the arithmetic mean of the distribution is 17.8.

2. The value of x is:

[CDS 2013 (1)]

- (a) 3 (b) 6
(c) 8 (d) 12

3. The median lies in which one of the following class intervals?

[CDS 2013 (1)]

- (a) 0-10 (b) 10-20

- (c) 20-30 (d) 30-40

4. The modal class is:

[CDS 2013 (1)]

- (a) 0-10 (b) 10-20
(c) 20-30 (d) 30-40

5. Consider the following statements:

- I. A frequency distribution condenses the data and reveals its important features
II. A frequency distribution is an equivalent representation of original data

Which of the above statement is/are correct?

[CDS 2013 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

6. Which one of the following statements is correct?

[CDS 2013 (1)]

- (a) A frequency polygon is obtained by connecting the corner points of the rectangles in a histogram
(b) A frequency polygon is obtained by connecting the mid points of the tops of the rectangles in a histogram
(c) A frequency polygon is obtained by connecting the corner points of the class intervals in a histogram
(d) None of the above

7. The average weight of a class of 15 boys and 10 girls is 38.4kg if the average weight of boys is 40 kg. Then what is the average weight of girls?

[CDS 2013 (1)]

- (a) 36.5kg (b) 35kg
(c) 36kg (d) 35.6kg

8. The average of u, v, w, x, y and z is 10. What is the average of $u + 10$, $v + 20$, $w + 30$, $x + 40$, $y + 50$ and $z + 60$?

[CDS 2013 (2)]

- (a) 30 (b) 35
(c) 40 (d) 45

9. If m is the mean of p, q, r, s, t and v, then what is $(p-m) + (q-m) + (r-m) + (s-m) + (t-m) + (v-m)$ equal to?

[CDS 2013 (2)]

- (a) 0 (b) s
(c) $\frac{(p+v)}{2}$ (d) None of these

Directions (next two): Read the following information carefully and answer the questions given below:

The median of the following distribution is 14.4 and the total frequency is 20.

Class Interval	0-6	6-12	12-18	18-24
	1	1	2	3
	2	8	4	0
Frequency	4	x	5	y
				1

10. What is x equal to?

[CDS 2013 (2)]

- (a) 4 (b) 5
(c) 6 (d) 7

11. What is the relation between x and y?

[CDS 2013 (2)]

- (a) $2x = 3y$ (b) $3x = 2y$
(c) $x = y$ (d) $2x = y$

12. Methods of presentation of data are:

[CDS 2013 (2)]

- (a) tables only (b) graphs only
(c) diagrams only (d) all of the above

13. Consider the following:

I. The arithmetic mean of two unequal positive numbers is always greater than their geometric mean

II. The geometric mean of two unequal positive numbers is always greater than their harmonic mean.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

14. Consider the following statements in respect of a discrete set of numbers:

I. The arithmetic mean uses all the data is always uniquely defined.

II. The median uses only one or two numbers from the data and may not be unique.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

15. The geometric mean of $(x_1, x_2, x_3, \dots, x_n)$ is x and the geometric mean of $(y_1, y_2, y_3, \dots, y_n)$ is y. Which of the following is/are correct?

I. The geometric mean of $(x_1, y_1, y_2, x_3, y_3, \dots, x_n, y_n)$ is XY.

II. The geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}\right)$ is $\frac{X}{Y}$.

Select the correct answer using the code given below:

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

16. The following table gives 'less than' type frequency distribution of income per day.

Income (in ₹) less than	Number of person
1500	100
1250	80
1000	70
750	55
500	32
250	12

What is the model class?

[CDS 2014 (1)]

- (a) 250-500 (b) 500-750
(c) 75-1000 (d) None of these

17. Which of the following items of information is a good example of statistical data?

[CDS 2014 (1)]

- (a) A table of logarithms of numbers
(b) A list of names of 120 students of a class
(c) A list of annual incomes of the members of a club
(d) Holiday list of the offices of Government of India in the year 2013.

18. Consider the following in respect of variate which takes value 2, 2, 2, 2, 7, 7, 7 and 7.

I. The median is equal to mean

II. The mode is both 2 and 7

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

19. Consider the following statements pertaining to a frequency polygon of a frequency distribution of a continuous variable having seven class intervals of equal width:

I. The original frequency distribution can be reconstructed from the frequency polygon

II. The frequency polygon touches the x-axis in its extreme right and extreme left.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

20. The mean of the following distribution is 18.

Class Interval	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

What is the value of f?

[CDS 2014 (1)]

- (a) 8 (b) 9
(c) 10 (d) 11

21. The class which has maximum frequency is known as:

[CDS 2014 (2)]

- (a) Median class (b) Mean class
(c) Modal class (d) None of these

22. Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range.

1. The cumulative frequency polygon gives an equivalent representation of frequency distribution table

2. The cumulative frequency polygon is a closed polygon with one horizontal and one vertical side

Which of the above statements is/are correct?

[CDS 2014 (2)]

- (a) Only 1 (b) Only 2
(c) Both I and 2 (d) Neither 1 nor 2

23. Consider the following data:

1. Number of complaints lodged due to road accidents in a state within a year for 5 consecutive years.

2. Budgetary allocation of the total available funds to the various items of expenditure

Which of the above data is/are suitable for representation of a pie diagram?

[CDS 2014 (2)]

- (a) Only 1 (b) Only 2
(c) Both I and 2 (d) Neither 1 nor 2

24. When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the area proportional to the frequencies of the respective class intervals, the graph so obtained is called:

[CDS 2014 (2)]

- (a) Bar diagram
(b) Frequency curve
(c) Ogive
(d) None of the above

25. If x_i 's are the mid points of the class intervals of grouped data, f_i 's are the corresponding frequencies at $\bar{x}$ is the mean, then what is $\sum f_i (x_i - \bar{x})$ equal to?

[CDS 2014 (2)]

- (a) 0 (b) -1
(c) 1 (d) 2

26. Ten observations 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32 and 43 are written in ascending order. The median of the data is 24. What is the value of x ?

[CDS 2014 (2)]

- (a) 15 (b) 18
(c) 20 (d) 24

27. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, then which one of the following relationship is correct?

[CDS 2015 (1)]

- (a) G is the geometric mean between A and H
(b) A is the arithmetic mean between G and H
(c) H is the harmonic mean between A and G
(d) None of the above

28. The geometric mean of three positive numbers a, b, c is 3 and the geometric mean of another three positive numbers d, e, f is 4. Also, at least three elements in the set (a, b, c, d, e, f) are distinct. Which one of the following inequalities gives the best information about M, the arithmetic mean of the six numbers?

[CDS 2015 (1)]

- (a) $M > 2\sqrt{3}$
(b) $M > 3.5$
(c) $M \geq 3.5$
(d) It is not possible to set any precise lower limit of M

29. There are five parties A, B, C, D and E in an election. Out of total 100000 votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allowed for party E in the pie chart?

[CDS 2015 (1)]

- (a) 15° (b) 54°
(c) 60° (d) 72°

For the next (4) items that follow:

Consider the following frequency distribution:

Class	Frequency
0-10	4
10-20	5
20-30	7
30-40	10
40-50	12
50-60	8
60-70	4

30. What is the mean of the distribution?

[CDS 2015 (1)]

- (a) 37.2 (b) 38.1
(c) 39.2 (d) 40.1

31. What is the median class?

[CDS 2015 (1)]

- (a) 20-30 (b) 30-40
(c) 40-50 (d) 50-60

32. What is the median of the distribution?

[CDS 2015 (1)]

- (a) 37 (b) 38
(c) 39 (d) 40

33. What is the mode of the distribution?

[CDS 2015 (1)]

- (a) 38.33 (b) 40.66
(c) 42.66 (d) 43.33

34. If a variable takes discrete values $a + 4$, $a - 3.5$, $a - 2.5$, $a - 3$, $a - 2$, $a + 0.5$, $a + 5$ and $a - 0.5$ where $a > 0$, then the median of the data set is:

[CDS 2015 (2)]

- (a) $a - 2.5$ (b) $a - 1.25$
(c) $a - 1.5$ (d) $a - 0.75$

35. If each of a n numbers $x_i = i$ ($i=1, 2, 3, \dots, n$) is replaced by $(i + 1) x_i$, then the new mean is:

[CDS 2015 (2)]

- (a) $\frac{n+3}{2}$ (b) $\frac{n(n+3)}{2}$

(c) $\frac{(n+1)(n+2)}{3n}$

(d) $\frac{(n+1)(n+2)}{3}$

36. The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to:

[CDS 2015 (2)]

- (a) 7 (b) 14
(c) 35 (d) 38.5

37. The election result in which six parties contested was depicted by a pie chart. Party A had an angle 135° on this pie chart. If it secured 21960 votes, how many valid votes in total were cast?

[CDS 2016 (1)]

- (a) 51240 (b) 58560
(c) 78320 (d) 87840

38. The mean and median of 5 observations are 9 and 8 respectively. If 1 is subtracted from each observation, then the new mean and the new median will respectively be:

[CDS 2016 (1)]

- (a) 8 and 7
(b) 9 and 7
(c) 8 and 9
(d) cannot be determined due to insufficient data

39. The age distribution of 40 children is as follows:

Age (in years)	5-	6-	7-	8-	9-	10-
					10	11
No. of Children	4	7	9	12	6	2

Consider the following statements in respect of the above frequency distribution.

- The median of the age distribution is 7 years
 - 70% of the children are in the age group 6-9 years
 - The modal age of the children is 8 years
- Which of the above statements are correct?

[CDS 2016 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

40. Suppose $x_i = \lambda^i$ for $0 \leq i \leq 10$, where $\lambda > 1$. Which one of the following is correct?

[CDS 2016 (1)]

- (a) $AM < \text{Median}$ (b) $GM < \text{Median}$
(c) $GM = \text{Median}$ (d) $AM = \text{Median}$

41. Suppose $x_i = \frac{1}{i}$ for $i = 1, 2, 3, \dots, 11$.

Which one of the following is not correct?

[CDS 2016 (1)]

- (a) $AM > 1/6$ (b) $GM > 1/6$
(c) $HM > 1/6$ (d) $\text{Median} = HM$

42. For $x > 0$, if a variable takes discrete values $x + 4, x - 3.5, x - 2.5, x - 3, x - 2, x + 0.5, x - 0.5, x + 5$, then what is the median?

[CDS 2016 (2)]

- (a) $x - 1.25$ (b) -0.5
(c) $x + 0.5$ (d) $x + 1.25$

43. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set:

[CDS 2016 (2)]

- (a) Is increased by 2
(b) Is decreased by 2
(c) Is two times the original median
(d) Remains the same as that of original set

44. Number of credit cards held by an individual can be treated as:

[CDS 2016 (2)]

- (a) Qualitative data
(b) Discrete data
(c) Categorical data
(d) None of the above

45. A pie chart depicts the classification of total funds of an organization according to different sources of funds. A particular sector of pie chart for corporate tax has 108° angle at the centre. What is the percentage of income from corporate tax to total funds?

[CDS 2016 (2)]

- (a) 20% (b) 25%
(c) 30% (d) 35%

46. Consider the following statements:

1. The classes of type 15-19, 20-24, 25-29 etc. are exclusive classes.

2. The classes of type 15-20, 20-25, 25-30 etc. are inclusive classes.

Which of the above statements is/are correct?

[CDS 2016 (2)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

47. Suppose the class interval 10-15 has frequency 30, then what is the frequency density of this class interval?

[CDS 2016 (2)]

- (a) 2 (b) 3
(c) 5 (d) 6

48. Consider the following distribution:

Class	Frequency
0-20	17
20-40	28
40-60	32
60-80	f
80-100	19

If the mean of the above distribution is 50, what is the value of f?

[CDS 2017 (1)]

- (a) 24 (b) 34
(c) 56 (d) 96

49. In a pie diagram, there are four slices with angles 150° , 90° , 60° and 60° . A new pie diagram is formed by deleting one of the slices having angle 60° in the given pie diagram. In the new pie diagram:

[CDS 2017 (1)]

- (a) The largest slice has angle 150°
(b) The smallest slice has angle 70°
(c) The largest slice has angle 180°
(d) The smallest slice has angle 90°

50. In an asymmetrical distribution, if the mean and median of the distribution are 270 and 220 respectively, then the mode of the data is:

[CDS 2017 (1)]

- (a) 120 (b) 220
(c) 280 (d) 370

51. An individual purchases three qualities of pencils. The relevant data is given below:

Quality Pencil	Price per pencil	Money spent (in Rs.)
A	1.00	50
B	1.50	X
C	2.00	20

It is known that the average price per pencil is Rs 1.25. What is the value of x?

[CDS 2017 (1)]

- (a) Rs 10 (b) Rs 30
(c) Rs 40 (d) Rs 60

52. Consider the following frequency distribution:

X	Frequency	Cumulative Frequency
1	8	8
2	10	18
3	f_1	29
4	f_2	45

What are the values of f_1 and f_2 respectively:

[CDS 2017 (2)]

- (a) 10 and 17 (b) 17 and 10
(c) 11 and 16 (d) 16 and 11

For the next five (05) items that follow:

In a University there are 1200 students studying four different subjects Mathematics, Statistics, Physics and Chemistry. 20% of the total number of students are studying Mathematics, one-fourth of the total number of students are studying Physics, 320 students are studying Statistics and remaining students are studying Chemistry. Three-fifth of the total number of students studying Chemistry are girls. 150 boys are studying Mathematics. 60% of student studying Physics are boys. 250 girls are studying Statistics.

53. What is the total number of boys studying Statistics and Physics?

[CDS 2017 (2)]

- (a) 180 (b) 240
(c) 250 (d) 310

54. The number of girls studying Statistics is what percentage (approximate) of the total number of students studying Chemistry?

[CDS 2017 (2)]

- (a) 58.8 (b) 73.5
(c) 78.7 (d) 80.6

55. In which subjects is the difference between the number of boys and girls equal to?

[CDS 2017 (2)]

- (a) Mathematics and Chemistry
(b) Statistics and Chemistry
(c) Mathematics and Physics
(d) Mathematics and Statistics

56. What is the difference between the number of boys studying Mathematics and the number of girls studying Physics?

[CDS 2017 (2)]

- (a) 20 (b) 30
(c) 60 (d) 80

57. What is the ratio of the total number of boys to the total number of girls?

[CDS 2017 (2)]

- (a) 67:83 (b) 17:26
(c) 27:19 (d) 189:179

58. Frequency density of a class is computed by the ratio:

[CDS 2017 (2)]

- (a) Class frequency to the class width
(b) Class frequency to total frequency
(c) Class frequency to of total number of classes
(d) Cumulative frequency up to that class to total frequency

59. The pie diagrams on the monthly expenditure of two families A and B are drawn with radii of two circles taken in the ratio 16:9 to compare their expenditure.

Which one of the following is the appropriate data used for the above mentioned pie diagram?

[CDS 2017 (2)]

- (a) Rs 16,000 and Rs 9,000
(b) Rs 8,000 and Rs 4,500
(c) Rs 25,600 and Rs 8,100
(d) Rs 4,000 and Rs 3,000

60. Consider the following statements:
Statement I: The value of a random variable having the highest frequency is mode
Statement II: Mode is unique
Which one of the following is correct in respect of the above statements?

[CDS 2017 (2)]

- (a) Both Statements I and II are true and statement II is the correct explanation of statement I.

(b) Both Statement I and II are true and statement II is not the correct explanation of statement I.

- (c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

61. Which one of the following is not correct?

The proportion of various items in a pie diagram is proportional to the

[CDS 2017 (2)]

- (a) Areas of slices
(b) Angles of slices
(c) Lengths of the curved arcs of the slices
(d) Perimeter of the slices

62. The geometric mean of x and y is 6 and the geometric mean of x , y and z is also 6. Then the value of z is:

[CDS 2017 (2)]

- (a) 12 (b) $\sqrt{6}$
(c) 6 (d) $3\sqrt{6}$

63. The total number of live births in a specific locality during different months of a specific year was obtained from the office the Birth Registrar. This set of data may be called:

[CDS 2017 (2)]

- (a) Primary data (b) Secondary data
(c) Recorded data (d) Countable data

64. The heights (in cm) of 5 students are 150, 161, 165, 144 and 155. What are the values of mean and median (in cm) respectively?

[CDS 2017 (2)]

- (a) 165 and 161 (b) 155 and 155
(c) 160 and 155 (d) 155 and 161

65. In the following table of inverse variation, what are the values of A , B and C respectively?

M	15	-6	2	C
N	-4	A	B	60

[CDS 2018 (1)]

- (a) 10, -30, -1 (b) 10, -1, 30
(c) -30, 10, -1 (d) -1, -30, -10

66. A Pie Chart is drawn for the following data:

Sector	Percentage
Agriculture and Rural Development	12.9
Irrigation	12.5
Energy	27.2
Industry and Minerals	15.4
Transport and Communication	15.9

Social Services

16.1

What is the angle (approximate) subtended by the Social Services Sector at the centre of the circle?

[CDS 2018 (1)]

- (a) 45° (b) 46°
(c) 58° (d) 98°

67. The arithmetic mean of two numbers is 10 and their geometric mean is 8. What are the two numbers?

[CDS 2018 (1)]

- (a) 15, 5 (b) 12, 8
(c) 16, 4 (d) 18, 2

68. If H is the harmonic mean of P and Q, then the value of $\frac{H}{P} + \frac{H}{Q}$ is:

[CDS 2018 (2)]

- (a) 1 (b) 2
(c) $\frac{P+Q}{PQ}$ (d) $\frac{PQ}{P+Q}$

69. Data on ratings of hotels in a city is measured on:

[CDS 2018 (2)]

- (a) Nominal scale (b) Ordinal scale
(c) Interval scale (d) Ratio scale

70. The median of 19 observations in 30. Two more observations are made and the values of these are 8 and 32. What is the median of the 21 observation?

[CDS 2018 (2)]

- (a) 32
(b) 30
(c) 20
(d) Cannot be determined due to insufficient data

71. As the number of observations and classes increases, the shape of a frequency polygon.

[CDS 2018 (2)]

- (a) Tends to become jagged
(b) Tends to become increasingly smooth
(c) Stays the same
(d) Varies only if data become more reliable

72. Consider the following statements:

Statement I: Median can be computed even when the end intervals of a frequency distribution are open.

Statement II: Median is a positional average. Which one of the following is correct in respect of the above statements?

[CDS 2018 (2)]

(a) Both Statements I and Statement II are true and Statement II is the correct explanation of statement I

(a) Both Statements I and Statement II are true and Statement II is the correct explanation of statement I

(b) Both Statements I and Statement II are true and Statement II is not the correct explanation of statement I.

(c) Statement I is true but statement II is false
(d) Statements I is false but statement II is true

Consider the following for the next three (03) items:

In a certain town of population size 1,00,000 three types of newspapers (I, II and III) are available. The percentages of the people in the town who read these papers are as follows:

Newspapers	Proportion of Readers
I	10%
II	30%
III	5%
Both I and II	8%
Both II and III	4%
Both I and III	2%
All the three (I, II, III)	1%

73. What is the number of people who read only one newspaper?

[CDS 2019 (1)]

- (a) 20,000 (b) 25,000
(c) 30,000 (d) 35,000

74. What is the number of people who read at least two newspapers?

[CDS 2019 (1)]

- (a) 12,000 (b) 13,000
(c) 14,000 (d) 15,000

75. What is the number of people who do not read any of these newspapers?

[CDS 2019 (1)]

- (a) 62,000 (b) 64,000
(c) 66,000 (d) 68,000

76. Consider the following grouped frequency distribution:

x	f
0-10	8
10-20	12
20-30	10
30-40	p
40-50	9

If the mean of the above data is 25.2, then what is the value of p ?

[CDS 2019 (1)]

- (a) 9 (b) 10
(c) 11 (d) 12

77. Consider the following frequency distribution:

x	f
8	6
5	4
6	5
10	8
9	9
4	6
7	4

What is the median for the distribution?

[CDS 2019 (1)]

- (a) 6 (b) 7
(c) 8 (d) 9

78. Diagrammatic representation of data includes which of the following:

1. Bar diagram
2. Pie diagram
3. Pictogram

Select the correct answer using the code given below:

[CDS 2019 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

79. The data collected from which one of the following methods is not a primary data?

[CDS 2019 (1)]

- (a) By direct personal interviews
(b) By indirect personal interviews
(c) By schedules sent through enumerators
(d) From published thesis

80. The monthly expenditure of a person is Rs 6,000. The distribution of expenditure on various items is as follows:

	Items of expenditure	Amount (in Rs)
1.	Food	2,000
2.	Clothing	660
3.	Fuel and rent	1200
4.	Education	480
5.	Miscellaneous	1660

If the above data is represented by a percentage bar diagram of height 15cm, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively?

[CDS 2019 (1)]

- (a) 1.25cm and 5cm
(b) 1.2cm and 4.15cm
(c) 1.2cm and 3.5cm
(d) 4.15 cm and 6cm

81. Which one of the following pairs is correctly matched?

[CDS 2019 (1)]

- (a) Median Graphical location
(b) Mean Graphical location
(c) Geometric mean Ogive
(d) Mode Ogive

82. The following pairs relate to frequency distribution of a discrete variable and its frequency polygon. Which one of the following pairs is not correctly matched?

[CDS 2019 (1)]

(a) Base line of the polygon	X-axis
(b) Ordinates of the vertices	Class frequency of the polygon
(c) Abscissa of the vertices	Class marks of The polygon frequency
(d) Area of the polygon	total frequency of the

Directions for the following two (02) items

Read the following frequency distribution for two series of observations and answer the two items that follow:

Class interval	Frequency	
	Series-I	Series-II
10-20	20	4
20-30	15	8
30-40	10	4
40-50	x	$2x$
50-60	y	y
total	100	100

83. What is the mean of frequency distribution of Series-I?

[CDS 2019 (2)]

- (a) 33-6 (b) 35-6
(c) 37-6 (d) 39-6

84. What is the mode of the frequency distribution of Series-II?

[CDS 2019 (2)]

- (a) 26 (b) 36
(c) 46 (d) 56

Directions for the following two (04) items

Read the following information and answer the four items that follow:

Let the distribution of number of scooters of companies X and Y sold 5 showrooms (A, B, C, D and E) in a certain year be denoted by S1 and the distribution of number of scooters of only company X and by the five showrooms in the same year be denoted by S2.

Show Room	A	B	C	D	E	Total number of scooters sold
S1 (in %)	1	2	1	5	1	6400
S2 (in %)	2	1	2	3	8	3000

85. Number of scooters of company Y sold by showroom E is what per cent of the number of scooters of both companies sold by showroom C?

[CDS 2019 (2)]

- (a) 52 (b) 54
(c) 55 (d) 56

86. Number of scooters of both the companies sold by showroom B is what per cent more than the number of scooters of company X and sold by showroom A?

[CDS 2019 (2)]

- (a) $78\frac{2}{3}$ (b) $83\frac{1}{3}$
(c) $86\frac{2}{3}$ (d) $88\frac{1}{3}$

87. What is the average number of scooters of company Y sold by the showroom A, C and E?

[CDS 2019 (2)]

- (a) $461\frac{1}{3}$ (b) $431\frac{1}{3}$
(c) $426\frac{1}{3}$ (d) $416\frac{1}{3}$

88. What is the difference between the number of scooters of both companies sold by showroom A and total number of scooters of company X sold by showroom B and E together?

[CDS 2019 (2)]

- (a) 416 (b) 426
(c) 432 (d) 436

Directions for the following four (04) items:

Read the following information and answer the four items that follow:

The data shows that Indian roads are turning deadlier over the year.

Year	2014	2015	2016	2017
Number of bikers killed	40957	46070	52750	48746
Number of pedestrian s killed	12330	13894	15746	20457
Number of cyclists killed	4037	3125	2585	3559

89. What was the average number of pedestrian killed per day in the year 2017?

[CDS 2019 (2)]

- (a) 51 (b) 53
(c) 54 (d) 56

90. What is the approximate percentage change in the pedestrians' fatalities during the period 2014-17?

[CDS 2019 (2)]

- (a) 66% (b) 68%
(c) 71% (d) 76%

91. What is the average number of bikers killed daily in road accidents in the year 2017?

[CDS 2019 (2)]

- (a) 163 (b) 152
(c) 147 (d) 134

92. What is the average number of cyclists killed daily in road accidents in 2017?

[CDS 2019 (2)]

- (a) 10 (b) 12
(c) 19 (d) 21

93. What is the median of the data 3, 5, 9, 4, 6, 11, 18?

[CDS 2020 (1)]

- (a) 6 (b) 6.5
(c) 7 (d) 7.5

94. In a pi-diagram there are three sectors. If the ratio of the angles of the sectors is 1:2:3, then what is the angle of the largest sector?

[CDS 2020 (1)]

- (a) 200° (b) 180°
(c) 150° (d) 120°

Directions for the following three (03) items:

Read the following information and answer the three items that follow:

The following data presents count of released convicts who have served prison terms (X), those who have received some educational or technical training during their term (Y) and those who were offered Company placement (Z) respectively, from six different jails A, B, C, D, E and F, in the year 2010.

	X	Y	Z
A	86	45	25
B	1305	903	461
C	2019	940	474
D	1166	869	416
E	954	544	254
F	1198	464	174

95. Jails with highest and smallest percentage of trained convicts are respectively:

[CDS 2020 (1)]

- (a) F and D (b) D and F
(c) C and A (d) D and A

96. Jail with highest placement rate of trained convicts is:

[CDS 2020 (1)]

- (a) F (b) D
(c) B (d) A

97. Jails from which more than half of the trained convicts are offered jobs, are:

[CDS 2020 (1)]

- (a) A, B and C (b) A, B and D
(c) A, D and E (d) A, E and F

98. For two observations, the sum of S and product is P. What is the harmonic mean of these two observations?

[CDS 2020 (1)]

- (a) $\frac{2S}{P}$ (b) $\frac{S}{(2P)}$
(c) $\frac{2P}{S}$ (d) $\frac{P}{(2S)}$

99. The maximum marks in a test are converted from 250 to 50 for the purpose of an Internal Assessment. The highest marks scored were 170 and lowest marks were 70. What is the difference between the maximum and minimum marks scored in the Internal Assessment.

[CDS 2020 (2)]

- (a) 15 (b) 17
(c) 20 (d) 24

100. Fifteen candidates appeared in an examination. The marks of the candidates who passed in the examination are 9, 6, 7, 8, 8, 9, 6, 5, 4 and 7. What is the median of marks of all the fifteen candidates?

[CDS 2020 (2)]

- (a) 6 (b) 6.5
(c) 7 (d) 7.5

101. If the yield (in gm) of barley from 7 plots of size one square yard each, were found to be 180, 191, 175, 111, 154, 141 and 176, then what is the median yield?

[CDS 2020 (2)]

- (a) 111gm (b) 154gm
(c) 175gm (d) 176gm

102. Which one of the following measures of central tendency will be used to determine the average size of the shoe sold in the shop?

[CDS 2020 (2)]

- (a) Arithmetic mean (b) Geometric
(c) Median (d) Mode

103. When the class intervals have equal width, the height of a rectangle in a histogram represents:

[CDS 2020 (2)]

- (a) Width of the class
(b) Lower class limit
(c) Upper class limit
(d) Frequency of the class

104. What is the algebraic sum of the deviations from the mean of a set of values 25, 65, 73, 75, 83, 76, 17, 15, 7, 14?

[CDS 2020 (2)]

- (a) -1 (b) 0
(c) 1 (d) 2

105. What is the median of 2, 4, 6, ..., 100?

[CDS 2020 (2)]

- (a) 48 (b) 49
(c) 50 (d) 51

- 106.** The harmonic mean and the geometric mean of two numbers are 10 and 12 respectively. What is their arithmetic mean?

[CDS 2020 (2)]

- (a) $\frac{25}{3}$ (b) $\sqrt{120}$
(c) 11 (d) 14.4

- 107.** Consider the following data with regard to production of cars (in lakhs):

	Year 2015	Year 2016
Country A	35	38
Country B	45	47
Country C	88	93
Country D	75	79
Country E	58	60.9

In which of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?

[CDS 2021 (1)]

- (a) B and E (b) A, C and D only
(c) A, C, D and E (d) A, D and E only

- 108.** The following table shows the marks of 90 students in a test of 80 marks:

Marks	Number of Students
1-10	5
11-20	8
21-30	10
31-40	13
41-50	18
51-60	17
61-70	12
71-80	7

The percentage of students who have obtained less than or equal to 50% marks is:

[CDS 2021 (1)]

- (a) 30% (b) 40%
(c) 45% (d) 60%

- 109.** What is the median of the following data?
2, 3, -1, 2, 6, 8, 9

[CDS 2021 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

Directions:

For the next four (04) items, consider the following data with regard to different types (I, II, III, IV, V) of multivitamin tables produced in a company (in lakhs)

Year	I	II	III	IV	V
2000	160	80	70	90	75

2001	200	150	85	160	100
2002	135	35	44	95	85
2003	240	95	120	80	120
2004	180	110	85	95	115
2005	210	150	100	92	110

- 110.** Which product is produced least over the years 2000-2005?

[CDS 2021 (1)]

- (a) Type II (b) Type III
(c) Type IV (d) Type V

- 111.** In which one of the following pairs of years, the difference in total number of tablets produced between them is minimum?

[CDS 2021 (1)]

- (a) (2003, 2005) (b) (2001, 2005)
(c) (2003, 2004) (d) (2000, 2002)

- 112.** The ratio of percentage drop in total production in 2004 compared to 2001 to that 2000 compared to 2001, is:

[CDS 2021 (1)]

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{5}$

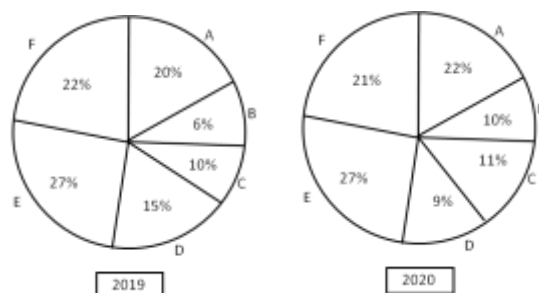
- 113.** In which year, the production of Type I is more than the sum of the production of Type II and Type IV?

[CDS 2021 (1)]

- (a) 2001 (b) 2002
(c) 2003 (d) 2004

Consider the following for the next four (04) items that follow:

The following Pie-Charts show the percentage of different categories (A, B, C, D, E and F) of employees in company in the year 2019 and 2020. The total number of employees was 4000 in 2019 and 5000 in 2020.



- 114.** There are two categories of employees in 2019 whose total strength remained the same in 2020. What are these two categories?

[CDS 2021 (2)]

- (a) C and B (b) E and A
(c) E and F (d) C and D

- 115.** In which one of the following categories of employees, the percentage change in number of employees in 2020 was maximum as compared to 2019?

[CDS 2021 (2)]

- (a) A (b) B
(c) C (d) D

- 116.** What was the percentage increase of category -E employees in 2020 as compared to that in 2019?

[CDS 2021 (2)]

- (a) 20% (b) 25%
(c) 30% (d) 40%

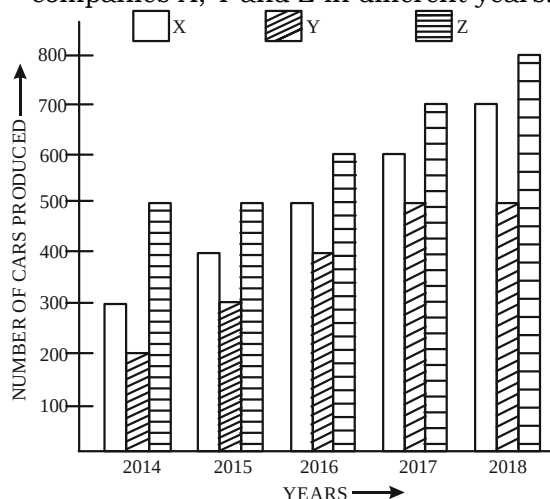
- 117.** There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories?

[CDS 2021 (2)]

- (a) E and F (b) C and E
(c) A and E (d) A and C

Consider the following for the next four (04) items that follows:

The following Bar-Chart gives the production of cars by three different companies X, Y and Z in different years:



- 118.** The percentage increase in the total production of cars from 2014 to 2018 was:

[CDS 2021 (2)]

- (a) 50% (b) 100%
(c) 150% (d) 200%

- 119.** Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20%?

[CDS 2021 (2)]

- (a) 5 (b) 6
(c) 7 (d) 8

- 120.** In which year was the percentage increase in production of cars minimum as compared to its previous year?

[CDS 2021 (2)]

- (a) 2015 (b) 2016
(c) 2017 (d) 2018

- 121.** In one of the years the percentage increase in production of cars was minimum as compared to its previous year. What was the minimum percentage?

[CDS 2021 (2)]

- (a) $11\frac{1}{9}\%$ (b) $11\frac{2}{9}\%$
(c) $13\frac{1}{3}\%$ (d) 20%

Direction: Consider the following data for the two (02) items that follows:

The table below gives the age-wise population percentage of a city:

Age group	Percentage
Below 30 years	14.00
30-34.99	29.75
35-39.99	26.25
40-44.99	0
45-49.99	18.50
50 years and above	11.50

The number of persons below the age of 40 years is 10.5 lakhs

- 122.** What is the total population of the city (in lakhs)?

[CDS 2022 (1)]

- (a) 21 (b) 18
(c) 15 (d) 12

- 123.** If the ratio of taxpayers to other persons in the same age group below 30 years is 1 : 2, then what is the number of taxpayers (in lakhs) in that age group?

[CDS 2022 (1)]

- (a) 0.4 (b) 0.7
(c) 0.85 (d) 1.05

Direction: Consider the following data for the two (02) items that follow:

The expenditure (in lakhs of rupees) of a company for the years 2011 to 2017 is as under:

Year	Expenditure
2011	13.8
2012	15.4
2013	10.4
2014	13.1
2015	15.8
2016	17.2
2017	19.4

- 124.** How many times the increase in expenditure in a year exceeded by more than 15% as compared to previous year?

[CDS 2022 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

- 125.** In which year, the percentage increase in expenditure is maximum as compared to its previous year?

[CDS 2022 (1)]

- (a) 2012 (b) 2014
(c) 2015 (d) 2017

Direction: Consider the following for the two (02) items that follow:

The budget allocations represented in a pie diagram under five different heads A, B, C, D and E are respectively 40%, 18%, 9%, 25% and 8%. The total budget allocation is ₹300.4 lakhs.

- 126.** How much less amount is allocated to A and C together as compared to B, D and E together?

[CDS 2022 (1)]

- (a) ₹3.004 lakhs (b) ₹4.005 lakhs
(c) ₹6.008 lakhs (d) ₹8.010 lakhs

- 127.** How much amount will be increased on A if the total budget is increased by three times?

[CDS 2022 (1)]

- (a) ₹360.48 lakhs (b) ₹300.36 lakhs
(c) ₹240.32 lakhs (d) ₹180.40 lakhs

- 128.** The arithmetic mean and the geometric mean of two positive numbers p and q ($p > q$) are A and G respectively. Which one of the following is correct?

[CDS 2022 (1)]

- (a) $A > G$ (b) $G > A$
(c) $A = G$ (d) $A^2 = G$

Consider the following for the next three (03) items that follow:

x	0	1	2	3	4	5
f	46	p	q	25	10	5

Total frequency is 200 and mean of the distribution is 1.46.

- 129.** What is the value of p?

[CDS 2022 (II)]

- (a) 70 (b) 72
(c) 76 (d) 78

- 130.** What is the value of q?

[CDS 2022 (II)]

- (a) 32 (b) 34
(c) 36 (d) 38

- 131.** What is the median of the distribution?

[CDS 2022 (II)]

- (a) 1 (b) 2
(c) 3 (d) 4

- 132.** If x is the harmonic mean between y and z, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $xy + xz - yz = 0$
(b) $xy + xz - 2yz = 0$
(c) $xy + xz + yz = 0$
(d) $xy + xz - 4yz = 0$

- 133.** What is the median of all possible factors of 120?

[CDS 2022 (II)]

- (a) 10 (b) 11
(c) 12 (d) 13.6

- 134.** The sum of deviations of a set of n values measured from 50 is -10 and the sum of deviations of the values measured from 46 is 70. What is the mean of the values?

[CDS 2022 (II)]

- (a) 48.5 (b) 49.0
(c) 49.5 (d) 50.0

- 135.** Consider the following table in respect of students of 4 schools who appeared in a test:

School	Name of students	Average marks in the test
I	60	60
II	50	80
III	50	40
IV	x	50

If the average marks of the students of all four schools are 58, then how many students appeared from School-IV?

[CDS 2022 (II)]

- (a) 38 (b) 40
(c) 42 (d) 44

- 136.** The data of different natural numbers 4, 7, 10, 14, $2x+3$, $2x+5$, 22, 23, 30, 50 are in ascending order. How many possible values are there for the median of the data for various values of x ?

[CDS 2022 (II)]

- (a) only one value
(b) only two values
(c) only three values
(d) five values

Consider the following data for the next two (02) items that follows:

Class	0-30	30-60	60-90	90-120
Frequency	4	5	7	4

- 137.** What is the mode of the distribution?

[CDS – 2023 (1)]

- (a) 60 (b) 72
(c) 75 (d) 80

- 138.** If the median (P) and mode (Q) satisfy the relation $7(Q-P) = 9R$, then what is the value of R?

[CDS – 2023 (1)]

- (a) 6 (b) 5
(c) 3 (d) 1

Consider the following data for the next two (02) items that follow:

Class	40-50	50-60	60-70	70-80
Frequency	4	3	1	2

- 139.** What is the mean of the distribution?

[CDS – 2023 (1)]

- (a) 51 (b) 52
(c) 54 (d) 56

- 140.** If M is the median, then what is the value $3M$?

[CDS – 2023 (1)]

- (a) $53\frac{1}{3}$ (b) 60
(c) 160 (d) 180

- 141.** The sum of deviations of n numbers from 10 and 20 are a , b respectively. If $\frac{b}{a} = -4$, then what is the mean of these n numbers?

[CDS-2023 (2)]

- (a) 12 (b) 14
(c) 16 (d) 18

- 142.** If the median of observations 12, 1, 8, 54, 61, 28, 45, 35, 21, 17 is M, then what is the value of $2M + 5$?

[CDS-2023 (2)]

- (a) 12 (b) 28
(c) 52 (d) 54

Consider the following for the next two (02) items that follow:

A grouped frequency distribution is given below:

Weekly wages in Rupees (₹)	Number of workers
2050-2550	5
2550-3050	10
3050-3550	k
3550-4050	8
4050-4550	2
4550-5050	10

- 143.** If average weekly wages earned by a worker is Rs.3520, then what is the value of k ?

[CDS-2023 (2)]

- (a) 10 (b) 12
(c) 15 (d) 20

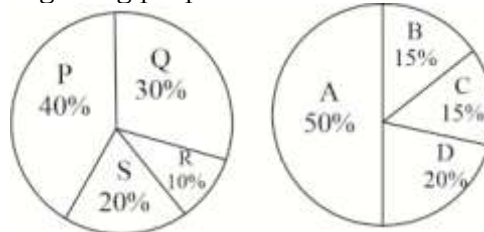
- 144.** What is the median (approximate value) of the distribution?

[CDS-2023 (2)]

- (a) Rs.3263 (b) Rs.3383
(c) Rs.3413 (d) Rs.3483

Consider the following for the next three (03) items that follow:

The following Pie-Chart-I shows the people migrating to Delhi from different Indian States (P, Q and R are three different States and S is the combined group of other States) and Pie-Chart-II indicates the different age groups A, B, C and D of these migrating people for each State.



- 145.** If the people coming from a particular State belong to S are 15% and 24,000 in number, then what is the total number of migrating people belonging to the age group B?

[CDS-2024 (1)]

- (a) 1.2 lac (b) 1.25 lac
(c) 1.30 lac (d) 1.50 lac .

146. What is the maximum of differences between the number of people coming from different groups P, Q, R and S?

[CDS-2024 (1)]

- (a) 1.6 lac (b) 1.8 lac
(c) 2.4 lac (d) 2.6 lac

147. What is the difference between number of people coming from R having age group A and those coming from Q having age group D?

[CDS-2024 (1)]

- (a) 6,000 (b) 8,000
(c) 12,000 (d) 18,000

148. Let the positive numbers $a_1, a_2, a_3, \dots, a_{3n}$ be in GP. If P is the GM of $a_1, a_2, a_3, \dots, a_n$ and Q is the GM of $a_{n+1}, a_{n+2}, a_{n+3}, \dots, a_{3n}$, then what is the GM of $3n$ numbers?

[CDS-2024 (1)]

- (a) P^2Q (b) PQ^2
(c) $\sqrt{PQ}$ (d) $P^{1/3}Q^{2/3}$

149. The sum of two positive numbers is 40. If the GM of these two numbers is lower than their AM by 20%, then what is the difference between the two numbers?

[CDS-2024 (1)]

- (a) 12 (b) 18
(c) 24 (d) 28

150. Consider the following data:

110, 41, 43, 95, 127, 99, 61, 92, 71, 93, 110, 36

If 93 is replaced by 94, then consider the following statements:

1. The difference between new median and old median is 1
2. The difference between new mean and old mean is less than 0.1
3. The difference between new mode and old mode is zero

Which of the statements given above are correct?

[CDS-2024 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ANSWER KEY

1.	b	2.	d	3.	b	4.	b	5.	c	6.	b	7.	c	8.	d	9.	a	10.	a
11.	b	12.	d	13.	c	14.	a	15.	c	16.	b	17.	d	18.	c	19.	c	20.	a
21.	c	22.	a	23.	c	24.	d	25.	a	26.	c	27.	a	28.	b	29.	b	30.	a
31.	b	32.	c	33.	d	34.	b	35.	d	36.	a	37.	b	38.	a	39.	c	40.	c
41.	c	42.	a	43.	d	44.	c	45.	c	46.	c	47.	d	48.	a	49.	c	50.	a
51.	b	52.	c	53.	c	54.	b	55.	c	56.	b	57.	a	58.	a	59.	a	60.	c
61.	d	62.	c	63.	b	64.	b	65.	a	66.	c	67.	c	68.	b	69.	b	70.	b
71.	b	72.	a	73.	a	74.	a	75.	d	76.	c	77.	c	78.	d	79.	d	80.	b
81.	a	82.	d	83.	c	84.	c	85.	c	86.	c	87.	a	88.	d	89.	d	90.	a
91.	d	92.	a	93.	a	94.	b	95.	b	96.	d	97.	a	98.	b	99.	c	100.	a
101.	c	102.	d	103.	d	104.	b	105.	d	106.	d	107.	c	108.	b	109.	b	110.	b
111.	a	112.	c	113.	c	114.	d	115.	b	116.	b	117.	d	118.	b	119.	a	120.	d
121.	a	122.	c	123.	b	124.	a	125.	b	126.	c	127.	c	128.	a	129.	c	130.	d
131.	b	132.	b	133.	b	134.	c	135.	b	136.	d	137.	b	138.	a	139.	d	140.	C
141.	a	142.	d	143.	c	144.	b	145.	a	146.	c	147.	b	148.	d	149.	c	150.	b